

Selecting The Right T&M Equipment For MEDICAL ELECTRONICS

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An overview of essential test parameters and technological improvements in test and measurement of medical devices

Proper maintenance of medical electronic devices is critically important. Medical equipment are expected to comply with many predefined standards, which require active T&M involvement to ensure accurate readings for remedy or diagnosis. All medical instruments—be it an ECG machine, an ultrasound machine or even a pacemaker—have their specific set of test devices. These devices are constantly improving by incorporating better technologies.

Optimal functioning of medical electronic devices depends on the environment, electrical capacity and operational processes. To know whether an equipment is being handled effectively, environmental parameters such as humidity, temperature, sound level, voltage, current and pH need to be measured. That is where basic T&M devices like thermometers, pH meters, voltmeters, ammeters and humid sensors come into play.

To improve user experience, test equipment are becoming smaller in size, shedding their bulky and space-consuming form factor. There are many more important upgrades in T&M equipment for medical devices, some of which are given below.

Signal testing

Electromedical hardware design is following the path of miniaturisation. Medical devices are increasingly using electronic chips with micro-actuators and micro-sensors. T&M has evolved accordingly by incorporating more test probes and going digital. Signal analysers and digital oscilloscopes are being modified with features fit for medical electronics.

Real-time functioning of medical devices is different from other electronic instruments. General-purpose devices can be tested by injecting known signals as test cases and verifying the output on an average to assure proper functioning in real life. But medical devices may need to monitor signals that vary every millisecond (like a person's heartbeat). Keeping this in mind, one of the most important features of medical T&M devices is their ability to measure single-shot signals.

Single-shot feature is essential to differentiate signal-to-noise ratio and get accurate readings. This ensures that the medical equipment being used is capturing correct signals under critical setups. Test setups for medical devices like ECG and EEG machines greatly benefit from these features.

T&M manufacturers are making their instruments more modular and compact. Amitesh Kumar, senior business development specialist, Liquid Instruments, cites an example: "One of the innovations in our test setup is an arbitrary waveform generator, which allows users to enter any equation including waveforms that can be analogous to the complex signal models generated by the human body in different test profiles. You need not have a computer with a catalogue full of different waveforms. This greatly speeds up report generation."

Electromagnetic compliance

Electromagnetic compliance (EMC) plays a crucial role in electromedical design and

